CARDIAC RISK NAVIGATOR

Explainable 3D Multi-Vessel CAD Decision Support System • Clinical Analysis Report

REPORT IDENTIFIER: CRN-20261011-BF6F7B
GENERATED: October 11, 2026 at 08:44:13
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Status: **Verified Model Inference**

Patient Identifier:	CRN-20261011-BF6F7B	**Age:**	**54** yrs	**Sex:**	**Male**
Height (Length):	**175** cm	**Weight:**	**80.5** kg	**BMI:**	**26.29** kg/m²

CLINICAL RISK STRATIFICATION SUMMARY: Multi-vessel decision support evaluation indicates an overall CAD model-predicted probability of **87.8%** (**HIGH RISK**). Individual vessel-level evaluation predicts LAD at **80.9%** (HIGH), LCX at **70.3%** (MODERATE), and RCA at **65.3%** (MODERATE). These figures represent outputs of trained machine-learning pipelines operating on 55 clinical parameters. They are decision-support estimates and do NOT reflect confirmed anatomic occlusion or physical percentage stenosis.

CAD — Overall Coronary Artery Disease

HIGH RISK

Significant multi-vessel CAD probability

87.8% (0.8778)

Tiers: LOW <45% | MOD 45-75% | HIGH ≥75%

Decision-Support Note: Model-predicted probability from verified pipeline. Not anatomical stenosis.

LAD — Left Anterior Descending Artery

HIGH RISK

Anterior wall & septal perfusion risk

80.9% (0.8089)

Tiers: LOW <45% | MOD 45-75% | HIGH ≥75%

Decision-Support Note: Model-predicted probability from verified pipeline. Not anatomical stenosis.

LCX — Left Circumflex Artery

MODERATE RISK

Lateral & posterior myocardial perfusion risk

70.3% (0.7029)

Tiers: LOW <45% | MOD 45-75% | HIGH ≥75%

Decision-Support Note: Model-predicted probability from verified pipeline. Not anatomical stenosis.

RCA — Right Coronary Artery

MODERATE RISK

Inferior & right ventricular perfusion risk

65.3% (0.6535)

Tiers: LOW <45% | MOD 45-75% | HIGH ≥75%

Decision-Support Note: Model-predicted probability from verified pipeline. Not anatomical stenosis.

DATA INTEGRITY & CLINICAL CAUTION: Patient demographics and risk metrics displayed above are derived solely from validated clinical inputs. No personal identifiers or measurements have been fabricated. Predicted probabilities are decision-support outputs and require clinical correlation with coronary angiography or CTA.

CORONARY ARTERY ANALYSIS & VESSEL STRATIFICATION

Machine-learning risk evaluation across individual epicardial coronary territories

CAD — Overall Coronary Artery Disease

Model: **RandomForestClassifier**

HIGH RISK

Perfusion Territory: Global Myocardial Perfusion (Epicardial Coronary Network)

87.8% (0.8778)

Clinical Context: Evaluates overall systemic likelihood of hemodynamically significant coronary atherosclerosis across any major vessel. Trained on comprehensive clinical, laboratory, and electrocardiographic features from the Z-Alizadeh Sani cohort.

Model Interpretation: A predicted probability of 87.8% places the patient in the HIGH risk category. This reflects model classification output based on statistical feature patterns. It does not quantify physical luminal diameter narrowing or substitute for invasive angiography.

LAD — Left Anterior Descending Artery

Model: **RandomForestClassifier**

HIGH RISK

Perfusion Territory: Anterior Left Ventricular Wall, Myocardial Apex & Interventricular Septum

80.9% (0.8089)

Clinical Context: The LAD typically perfuses 45%–55% of left ventricular myocardial mass. Stenosis in this vessel carries high prognostic significance ('widow-maker' territory). Features like anterior ECG changes, chest pain types, and RWMA strongly influence this pipeline.

Model Interpretation: LAD model probability is 80.9% (HIGH risk). This probability represents classifier confidence for LAD stenosis presence. It does NOT establish confirmed stenosis severity, lesion length, or exact anatomical plaque location.

LCX — Left Circumflex Artery

Model: **XGBClassifier (Extreme Gradient Boosting)**

MODERATE RISK

Perfusion Territory: Posterolateral Left Ventricle & Obtuse Marginal Territories

70.3% (0.7029)

Clinical Context: The LCX travels in the coronary sulcus, supplying lateral and posterior ventricular walls. LCX lesions can be electrocardiographically silent on standard 12-lead ECG. The XGBoost model utilizes gradient-boosted decision trees optimized for this target.

Model Interpretation: LCX model probability is 70.3% (MODERATE risk). Output reflects tree-based non-linear risk interactions. It is a diagnostic aid, not physical artery caliper measurement or proof of anatomical stenosis.

RCA — Right Coronary Artery

Model: **RandomForestClassifier**

MODERATE RISK

Perfusion Territory: Right Ventricle, Inferior Left Ventricle & SA/AV Nodal Conduction System

65.3% (0.6535)

Clinical Context: In right-dominant anatomy (~85% of individuals), RCA supplies the posterior descending artery (PDA) and inferior septum. Inferior lead findings (II, III, aVF) and hemodynamic parameters contribute prominently.

Model Interpretation: RCA model probability is 65.3% (MODERATE risk). Indicates statistical likelihood of significant right coronary involvement. It is not a confirmed stenosis finding and must be validated through standard clinical imaging.

CLINICAL DATA SUMMARY

Validated clinical features utilized for model inference. Missing items are designated 'Not provided'.

A. DEMOGRAPHICS

Age	**54** years	**Sex**	**Male**
Height	**175** cm	**Weight**	**80.5** kg
BMI	**26.29** kg/m²		

B. MEDICAL HISTORY

Diabetes (DM)	**1**	**Hypertension (HTN)**	**0**
Current Smoker	**1**	**Former Smoker**	*Not provided*
Family History (FH)	*Not provided*	**Obesity**	*Not provided*
Chronic Renal Failure	*Not provided*	**Stroke / CVA**	*Not provided*
Airway Disease	*Not provided*	**Thyroid Disease**	*Not provided*
Heart Failure (CHF)	*Not provided*	**Dyslipidemia (DLP)**	*Not provided*

C. SYMPTOMS & CLINICAL EXAMINATION

Typical Chest Pain	**1**	**Atypical Angina**	*Not provided*
Non-anginal Pain	*Not provided*	**Exertional CP**	*Not provided*
Low-Threshold Angina	*Not provided*	**Dyspnea**	*Not provided*
NYHA Functional Class	*Not provided*	**Blood Pressure**	**120** mmHg
Pulse Rate	**72** bpm	**Edema**	*Not provided*
Weak Peripheral Pulse	*Not provided*	**Lung Rales**	*Not provided*
Systolic Murmur	*Not provided*	**Diastolic Murmur**	*Not provided*

D. ELECTROCARDIOGRAPHY (ECG)

Q Wave	*Not provided*	**ST Elevation**	*Not provided*
ST Depression	*Not provided*	**T-Wave Inversion**	*Not provided*
Left Ventricular Hypertrophy	*Not provided*	**Poor R Progression**	*Not provided*
Bundle Branch Block	*Not provided*		

E. LABORATORY RESULTS

Fasting Blood Sugar	**105** mg/dL	**Serum Creatinine**	**1.1** mg/dL
Triglycerides	**150** mg/dL	**LDL Cholesterol**	**110** mg/dL
HDL Cholesterol	**45** mg/dL	**Blood Urea Nitrogen**	**15** mg/dL
Erythrocyte Sed. Rate	**12** mm/hr	**Hemoglobin**	**14.5** g/dL
Potassium (K)	**4.2** mEq/L	**Sodium (Na)**	**138** mEq/L
White Blood Cell Count	**7500** /µL	**Lymphocytes**	**30** %
Neutrophils	**60** %	**Platelet Count**	**250** ×10³/µL

F. ECHOCARDIOGRAPHY

Ejection Fraction (EF)	**50** %	**Regional RWMA (Score)**	*Not provided*
Valvular Heart Disease	*Not provided*		

EXPLAINABLE AI: FEATURE ATTRIBUTION ANALYSIS (SHAP)

Patient-specific Shapley Additive exPlanations quantifying clinical drivers of predicted risk

SHAP METHODOLOGY & SCALE INTERPRETATION: Tree-based SHAP attributions quantify how much each clinical parameter shifts the prediction from the cohort baseline. **Positive (+) values** indicate features that elevate predicted risk, while **negative (-) values** indicate protective or risk-attenuating features.
Scale Difference Notice: Random Forest models (CAD, LAD, RCA) calculate attributions on the **probability scale** (pts). The XGBoost model (LCX) calculates attributions on the **log-odds margin scale**. Due to differing output scales, raw SHAP values should not be compared directly across different target models.

CAD — Overall CAD Risk Attribution (Random Forest Pipeline)

Scale: **Probability Scale (+pts)**

Clinical Feature	SHAP Value	Direction	Impact Description
Typical Chest Pain	**+0.1520** pts	**Elevates Risk (+)**	Pushes model output toward stenosis by 0.1520 pts
Age	**+0.0810** pts	**Elevates Risk (+)**	Pushes model output toward stenosis by 0.0810 pts

LAD — LAD Artery Risk Attribution (Random Forest Pipeline)

Scale: **Probability Scale (+pts)**

Clinical Feature	SHAP Value	Direction	Impact Description
Typical Chest Pain	**+0.1210** pts	**Elevates Risk (+)**	Pushes model output toward stenosis by 0.1210 pts
EF-TTE	**-0.0450** pts	**Reduces Risk (-)**	Pushes model output toward normal by 0.0450 pts

LCX — LCX Artery Risk Attribution (XGBoost Pipeline)

Scale: **Log-Odds Margin Scale**

Clinical Feature	SHAP Value	Direction	Impact Description
Age	**+0.4320** log-odds	**Elevates Risk (+)**	Pushes model output toward stenosis by 0.4320 log-odds
DM	**+0.3120** log-odds	**Elevates Risk (+)**	Pushes model output toward stenosis by 0.3120 log-odds

RCA — RCA Artery Risk Attribution (Random Forest Pipeline)

Scale: **Probability Scale (+pts)**

Clinical Feature	SHAP Value	Direction	Impact Description
Age	**+0.0650** pts	**Elevates Risk (+)**	Pushes model output toward stenosis by 0.0650 pts
HTN	**-0.0320** pts	**Reduces Risk (-)**	Pushes model output toward normal by 0.0320 pts

3D ANATOMICAL RISK VISUALIZATION & BENCHMARK VALIDATION

Approximate vessel-level risk representation and previously reported model evaluation metrics

3D Coronary Vessel Risk Mapping

LAD: **HIGH** | LCX: **MODERATE** | RCA: **MODERATE**

3D heart model rendering reflects patient-specific vessel risk levels mapped onto the coronary arterial tree. In interactive mode, vessels are highlighted based on model risk categories.

APPROXIMATE RISK REPRESENTATION NOTICE: The 3D heart visualization is an approximate computational representation of model-predicted vessel-level risk. It does **NOT** identify exact plaque location, calcified lesion morphology, or actual physical stenosis severity. It must not be interpreted as anatomical CT angiography or catheterization imaging.

MODEL BENCHMARK PERFORMANCE (FIVE-FOLD CROSS-VALIDATION)

Previously established validation metrics on the research cohort (Z-Alizadeh Sani dataset, 303 records).

Target Territory	Trained Pipeline	Accuracy	F1-Score	ROC-AUC	Validation Protocol
CAD	Random Forest	87.78%	91.32%	92.79%	5-Fold Cross-Validation
LAD	Random Forest	80.89%	84.10%	86.13%	5-Fold Cross-Validation
LCX	XGBoost	70.29%	57.59%	74.25%	5-Fold Cross-Validation
RCA	Random Forest	65.35%	54.30%	71.87%	5-Fold Cross-Validation

REPORTED METRIC CONTEXT: These evaluation figures represent previously reported 5-fold cross-validation results on the retrospective research cohort. They do NOT reflect the diagnostic accuracy of this specific patient's prediction, nor do they represent independent external clinical trial validation.

MEDICAL SAFETY & LIMITATIONS DISCLAIMER:
Research and Educational Decision-Support Prototype.
The probabilities in this report are outputs of machine-learning models and are not confirmed clinical diagnoses or measurements of physical artery blockage.
The 3D visualization is an approximate representation of predicted vessel-level risk and does not establish exact plaque location or anatomical stenosis severity.
This report is not a substitute for clinical evaluation, coronary imaging, or professional medical judgment. No treatment plans, prescriptions, or invasive interventions should be initiated based solely on this document.